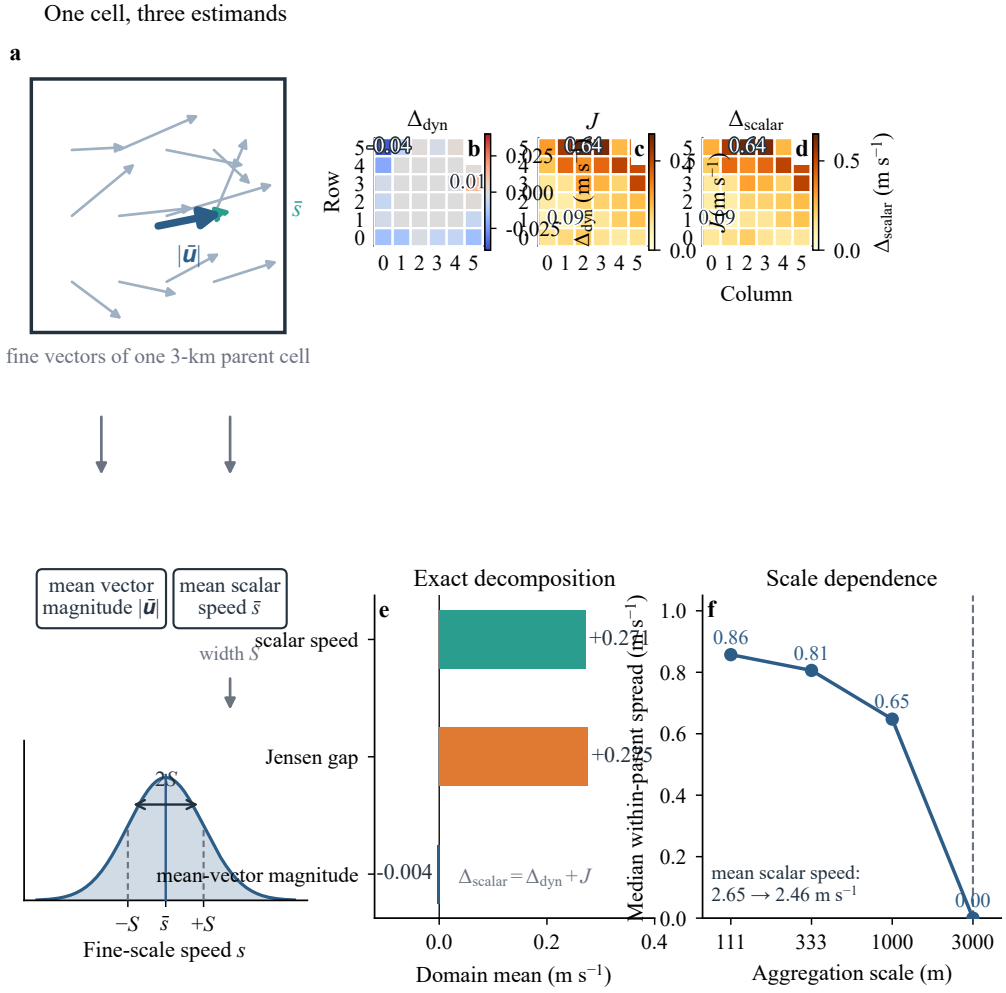

One cell, three estimands
a
$\bar{s}$
$|\bar{\boldsymbol{u}}|$
fine vectors of one 3-km parent cell
Δ_{dyn}
J
Δ_{scalar}
b
c
d
-0.04
0.01
0.64
0.09
0.64
0.09
Row
Column
Δ_{dyn} (m s^{-1})
J (m s^{-1})
Δ_{scalar} (m s^{-1})
mean vector magnitude $|\bar{\boldsymbol{u}}|$
mean scalar speed $\bar{s}$
width S
$2S$
mean-vector magnitude
$-S$
$\bar{s}$
$+S$
Fine-scale speed s
Exact decomposition
e
scalar speed
Jensen gap
+0.211
+0.225
-0.004
$\Delta_{scalar} = \Delta_{dyn} + J$
Domain mean (m s^{-1})
Median within-parent spread (m s^{-1})
Scale dependence
f
0.86
0.81
0.65
0.00
mean scalar speed:
2.65 → 2.46 m s^{-1}
Aggregation scale (m)